

CYBER THREATS AND THEIR MITIGATION



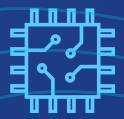
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PROBLEM STATEMENT

With the advent of information technology, the threat of cybersecurity attacks have increased at an exponential rate in the last decade. Cyber threats come in various forms and deceive millions each year, including the technologically adept. Cyber crimes result in significant financial losses for individuals and businesses, an estimated \$23 trillion by 2027. Mitigations for these threats must be thought at institutional, individual and technological levels.

EXISTING WORKS AND LIMITATIONS



POST QUANTUM ENCRYPTION STANDARDS

Advances in quantum computing threaten widely deployed asymmetric cryptographic systems.

To address this risk, the National Institute of Standards and Technology (NIST) has standardized four post-quantum cryptographic algorithms: CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON, SPHINCS.

Even though these standards have been made public, a lot of security research needs to be done before they are deployed for public use.



AI-POWERED THREAT DETECTION

AI systems are trained to recognize unusual behaviour in systems to detect DoS and zero-day attacks in real-time. Similarly, it can also detect malware based on the system activity. NLP models help detect suspicious language patterns in phishing and fraud attacks.

These models often provide accurate results but lack transparency. They are also prone to adversarial attacks.



CONCLUSIONS

Cyber threats continue to grow in scale, sophistication, and impact. Defensive strategies must evolve at the same pace as emerging technologies. Implementing quantum-resistant cryptography, strengthening system architectures, and maintaining continuous vigilance are essential steps. Long-term security depends on preparedness, adaptability, and sustained investment in resilient digital infrastructure.

REFERENCES

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DIFFERENT FORMS OF CYBER THREATS



PHISHING

Attackers masquerade as reputable entities to trick victims into revealing sensitive information



IDENTITY THEFT

A crime where attacker uses another person's identifying information without permission to commit crimes.



MALWARE

Malicious Software intentionally designed to disrupt, steal data, damage, or gain unauthorized access to systems.



RANSOMWARE

A form of malware where attackers use ransomware as a tool to steal data and hold it hostage until payment received



DENIAL OF SERVICE

Attempt to disrupt a targeted network by overwhelming it with a flood of Internet traffic, making it unavailable to users

MITIGATIONS



Training Programs and Multi Factor Authentication systems



Strong passwords, encryption and identity verification



Antivirus solutions and regular security patches



Offline backups and least-privilege access controls



DDoS protection and intrusion prevention systems



ISO 27001 for information security management



Reporting to CERT-IN and following advisories